Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Webmobix - RMSignal

Wealth Management Intelligence.
Move from portfolio monitoring to proactive, explainable advice for Relationship Managers.

Brief about the idea

The problem

A Relationship Manager (RM) has 20 clients and USD 596m to look after. Today's tools show numbers. They do not say who to call first or why.

What RMSignal does

It reads portfolios, mandates, market events and the RM's own notes. It gives back a ranked list of next actions: who to call, what to say and the proof behind it.

Who stays in charge

The RM. She can inspect, question and overrule everything. The AI suggests. She decides.

RMSignal turns portfolio data into a short, ranked, explained to-do list, so the Relationship Manager knows who to call first, what to say, and why, with the proof one click away.

Opportunities

How is it different?

- Most tools describe one portfolio. RMSignal ranks actions across the whole book.
- It looks across a client's portfolios and inside structured products.
- It reads the RM's own notes and flags where they disagree with the data.

How does it solve the problem?

- Code finds mandate breaches, concentration, cash gaps, loan limits and overdue KYC.
- Actions are ranked by importance and timing, so the RM knows who to call first.
- Three scenarios show best, likely and worst cases for each client.

What is the USP?

- Explainable: one click from any claim to its source data.
- Safe: the AI never does the maths, and code deletes any claim whose numbers are not in the evidence.
- Honest: every report lists what could not be checked.

List of features offered by the solution

Ranked action list

Every action for every client, sorted by importance and timing. A written summary says who to call first.

Client view

All portfolios together: allocation, currencies, biggest positions and what structured products really hold.

Notes against data

Each RM note is checked: consistent, contradicted or cannot be verified.

Mandate breaches

Code finds the breach. Notes say if it is acknowledged, client-directed or contradicted. A waiver never hides it.

Cash and credit

Liquid assets against planned spending, loan-to-value against limits, KYC review dates.

Scenarios

Gulf de-escalation, Gulf deterioration, Rates higher for longer. Best, likely and worst cases.

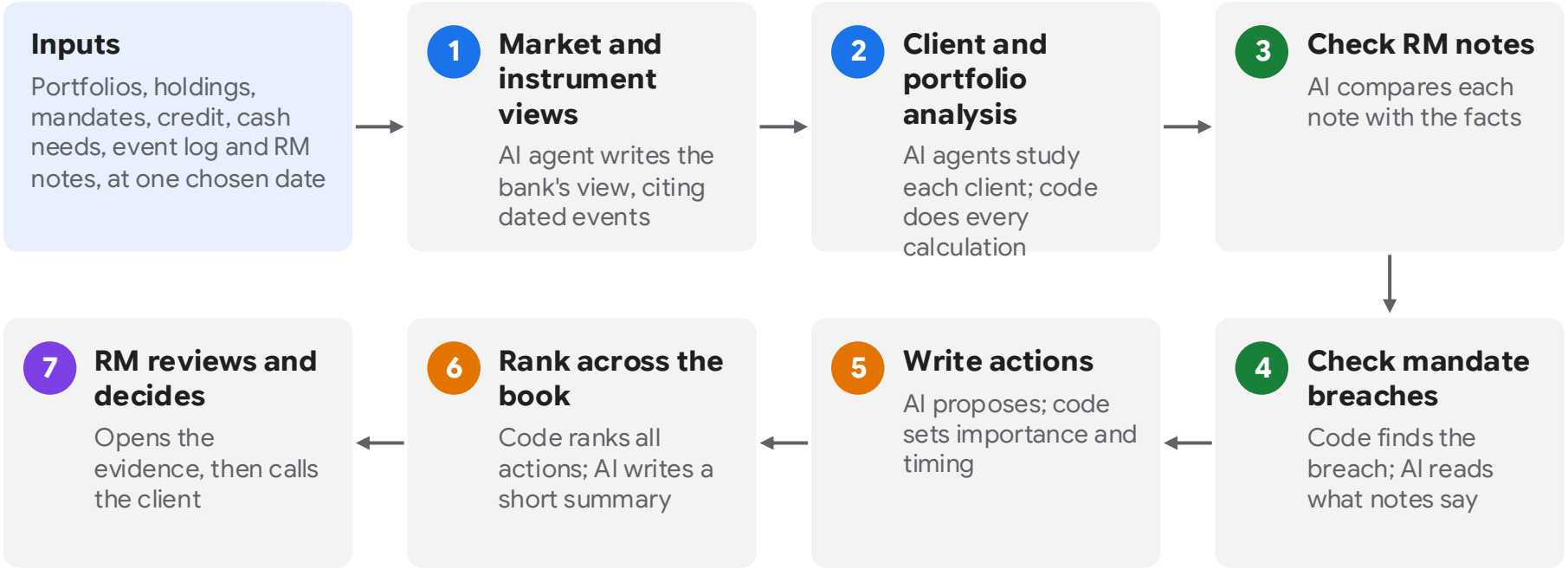
Bank views

Market and instrument views, each tied to dated events from the event log.

Evidence drawer

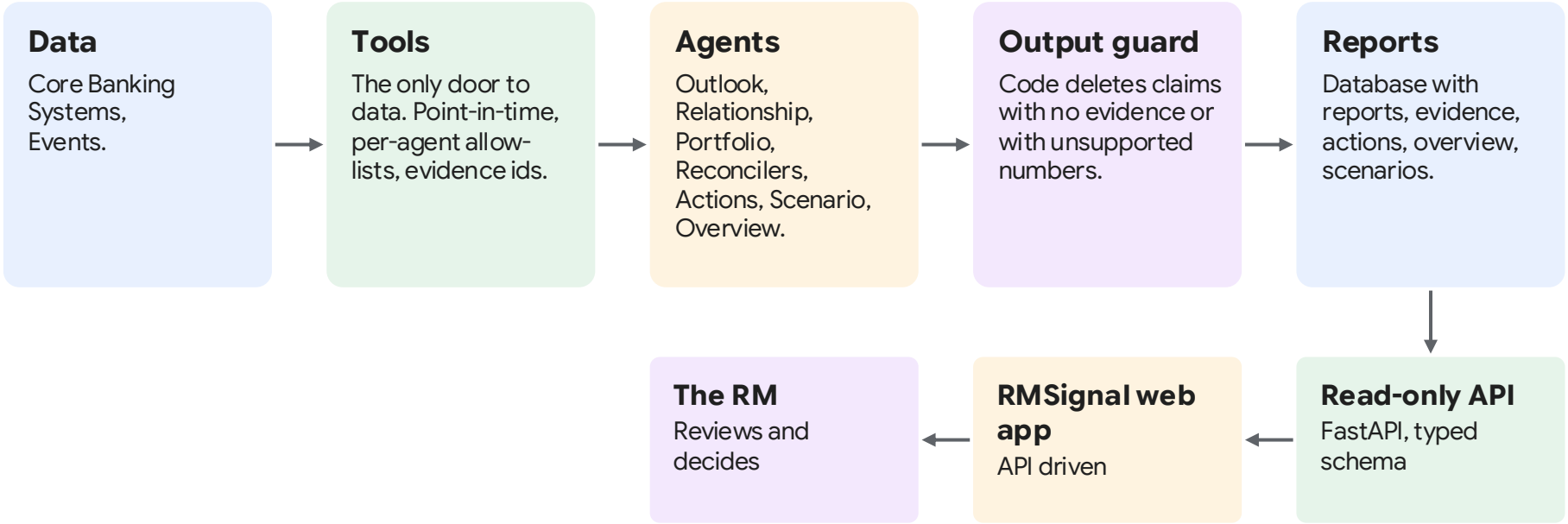
Click any claim to see the data, the figures and the note behind it.

Process flow diagram or Use-case diagram



Each step saves a file and the next step reads it. The same date and the same inputs give the same result.

Architecture diagram of the proposed solution



Across every layer: policy checkpoints (mock engine) | audit log of every tool call | client names replaced before any model call | one fixed date per run, no look-ahead | KYC peer over A2A

Technologies to be used in the solution

AI and agents

- Deep Agents
- LangGraph
- LangChain
- Any OpenAI-compatible model
- A2A (agent to agent) SDK

Data and analytics

- Python 3.13
- Polars

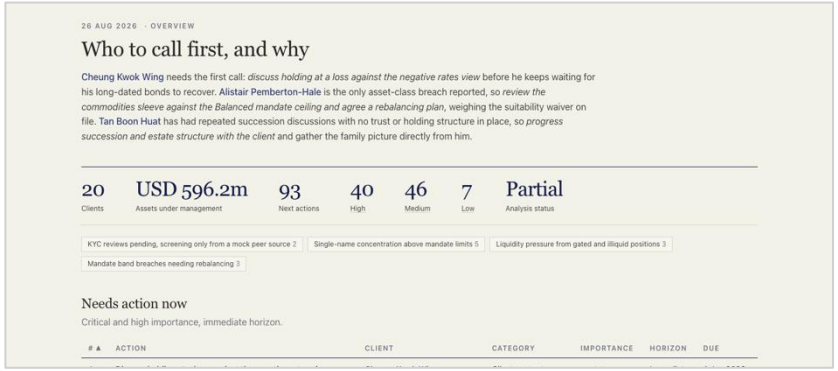
Dashboard

- TanStack Start
- React 19
- TanStack Router
- TanStack Query
- TanStack Table
- TanStack Charts
- Tailwind CSS 4
- Vite

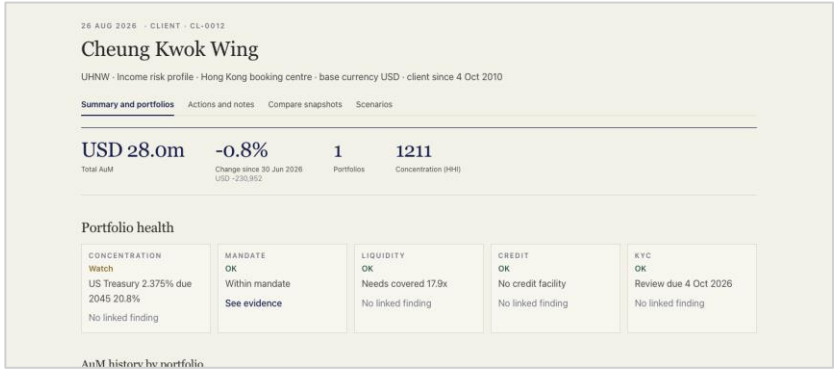
Quality

- pytest
- Hypothesis
- Vitest
- Biome
- OpenAPI-generated types

Snapshots of the prototype



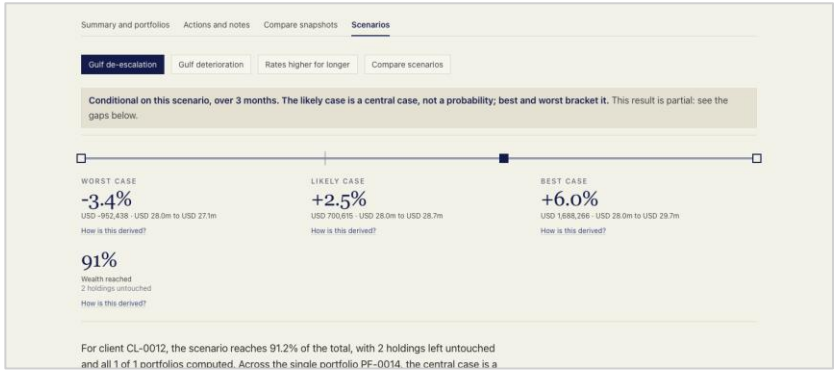
Overview: who to call first, and why



Client page: portfolio health at a glance



RM notes checked against the data



Scenarios: worst, likely and best case

Prototype Performance report/Benchmarking

846+
backend tests pass

100
dashboard tests pass (27 files)

20 of 20
clients analysed in one run

161
model calls in that run, 0 failed

10 min
to analyse the whole book, 4 clients at a time

2.2m
tokens used for the full analysis

What these numbers mean

- Reports are repeatable: the same date and inputs give the same result, and unchanged outlooks cost no model call.
- All 20 client reports are marked partial on purpose. The dataset has no related persons, and the report says so.
- The guard works live: on the client page, a finding was removed because a figure was not in its evidence.
- This is a prototype run on 20 synthetic clients, not a production load test.

Additional Details/Future Development (if any)

Also built

- Interactive analyst sessions for follow-up questions
- Agent-to-agent (A2A) cards and servers
- A model-call log with counts and timing, never content
- Point-in-time engine: no data after the chosen date is ever used
- Call sheet per portfolio: seven questions an RM is asked in a meeting

Next steps

- Connect the bank's real policy engine and KYC system
- Add related persons and beneficial owners when data exists
- Add an external news feed behind the same tool
- Add tax-aware ideas and life-event planning
- Let the RM accept, reject or edit each action
- Add sign-in and per-RM access

Provide links to your:

GitHub Public Repository
https://github.com/webmobix/rmsignal

Demo Video Link (3 Minutes)
[add link before submitting]

Final Product Link
https://rm-dashboard-spa-820913152799.asia-southeast3.run.app/

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Thank you

